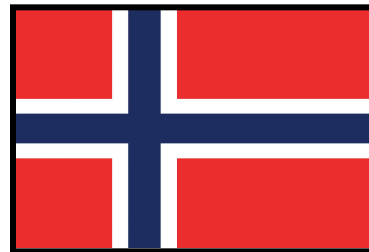


The Efficacy² of Renewable Energy Policy on reducing CO2 emissions and Economic Development

GROUP A INDIA & NORWAY



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I. INTRODUCTION

Rising global emissions are causing enormous damage to the earth, owing primarily to the combustion of fossil fuels and deforestation. According to the International Energy Agency (IEA, 2023), worldwide CO₂ emissions will reach 36.8 gigatons in 2022, the highest level ever recorded, a 0.9% rise from 2021. This increase in emissions has exacerbated climate change, making severe weather events like heat waves, floods, and droughts more common and catastrophic. These consequences endanger ecosystems, coastal regions, and human health, making effective renewable energy policy more important than ever. Renewable energy sources such as wind, solar, and hydroelectric power offer a sustainable alternative to fossil fuels while generating much less greenhouse gas.

Renewable energy is a source of energy in which resources are naturally replenished, such as sun, wind, water, or geothermal heat. Unlike fossil fuels, these sources are virtually endless and clean, and are the source of global carbon emissions that are vital to curb. First fossil fuels pollute the environment and they are also finite and ecosystem destructive. On the other hand, renewables are sustainable thus offering a solution to the growing threat of climate change as they lower the world's carbon footprint. (Banton, 2024)

Renewables energy policies commenced in the 1970s when the oil crises emphasized the need to be independent in energy. Meanwhile, early investments of solar and wind power were made during this period. But these policies didn't gain traction until the 1990s and early 2000s as the policy frameworks matured. Countries like Germany created feed-in tariffs (FITs), the U.S. had renewable portfolio standards (RPS) that mandated utilities to create a percentage of their energy from renewables. The Kyoto Protocol (1997), which set greenhouse gas emission targets for industrialized nations, and the Paris Agreement (2015), which set a goal to reduce global temperature rise this century, also prompted those like China and Poland to act. The global accords act as a booster to the case of renewable energy, pressuring governments to support clean energy technologies and setting ambitious carbon reduction goals. During the 2010s, falling prices for wind and solar energy, and technological advancement, made renewable sources cheaper. Meanwhile, countries started to set carbon neutral goals and moved away from subsidies to market based methods including auctions and carbon markets in order to promote new economies in using renewables. (Gomstyn, 2024).

Importance of Renewable Energy Policies

Currently, renewable energy policies aim to incorporate technologies like energy storage and green hydrogen. The ability of these new innovations is crucial for developing future-proof, decarbonized energy systems that can replace fossil fuels with reliability. Strong renewable energy policies are imperative for facilitating a transition from fossil fuel to low-carbon energy systems (REN21, 2021). Without such policies, the renewable energy industry would find itself at a serious disadvantage in the face of fossil fuel sectors that receive deep subsidies while enjoying heavily entrenched infrastructure (Bruckner et al., 2014). These policies are also necessary to deal with climate change. Renewable energy policies are, therefore, encouraging the use of cleaner, lower carbon alternatives, as global temperatures rise amid the burning of fossil fuels (IPCC, 2021).

Governments can lessen carbon emissions and slow the rate of global warming by promoting the use of renewables. Furthermore, renewable energy policies increase the security of energy supply by expanding energy access and reducing dependence on volatile fossil fuel markets for those countries that rely mostly on foreign oil and gas imports (IRENA, 2019). Renewable energy policies also offer economic development and job creation benefits, in addition to environmental ones. When it comes to clean energy, the transition creates new opportunities in the manufacture of solar panels, installation of wind turbines, and the modernisation of power grids. While clean energy technologies can drive innovation and spur private sector investment in new industries, the jobs created by these new industries can also stimulate local economies (Bruckner et al., 2014). Furthermore, with more availability of renewable energy, it can aid in reducing energy inequality by bringing clean, low-cost energy to underserved communities, thereby supporting social justice with environmental sustainability (REN21, 2021).

Among the most serious threats to human health and the environment is the burning of fossil fuels, which releases carbon into the atmosphere. The trapping of heat by increasing amounts of carbon dioxide, methane, and nitrous oxide in the atmosphere is a process called global warming, and leads to global temperatures rising (IPCC, 2021). This warming effect produces extreme weather events like floods, hurricanes, and droughts, devastating not only natural life and human communities, but their very ecosystems. Another way the rise in carbon emissions contributes to more frequent, more intense climate-related disasters. Damage to communities, economies, and at-risk lives is coming from stronger storms, longer droughts, and more frequent wildfires. Melting polar ice and glaciers will raise sea levels, which will displace millions of people living in coastal areas and lower-lying areas. These changes are already having real impacts on food production, water availability, and public health on a global scale (UNFCCC, 2015).

As well as overall carbon emissions, other factors are involved in the degradation of ecosystems. Ocean acidification is the process by which the oceans absorb excess carbon dioxide and harm marine life, most specifically coral reefs. In turn, it affects biodiversity and the health of entire marine ecosystems. In many cases, changing temperatures and rainfall patterns on land are threatening the habitats of hundreds of species, many of which can't adapt as quickly as they need in order to survive (Bruckner et al., 2014). Carbon emissions have broad health impacts. Respiratory and cardiovascular diseases are caused by air pollution from burning fossil fuels, taking millions of lives per year (WHO, 2021). Warmer temperatures are also increasing the range of vector-borne diseases such as malaria and dengue fever, putting new health risks on populations that have not been burdened with these diseases before.

The importance of reducing carbon emissions is as critical to mitigating these negative impacts in the context of administrative energy policies. Enabling governments to shift from using fossil fuels to clean energy sources such as wind, solar, and hydropower is a significant way to decrease their carbon footprint and contribute to a cleaner, more sustainable future. The renewable energy transition is not just an environmental need; it's an economic and social challenge to create a more resilient and nonpartisan world (UNFCCC, 2015).

Country Profiles

1. India

India is one of the largest carbon emitters among the list of countries in the world, being the third-largest in total emissions of CO₂, especially because of the heavy dependence on coal for its electricity generation. With a fast-developing society and increasing population, which means a sudden rise in energy consumption over the years, India qualifies as the second-fastest expanding economy (IRENA, 2019). The major contributors of carbon emissions in the country include industries, power generation, and transportation, with coal being the primary energy earner, covering about 70% of the electricity generation. However, this contributes greatly to the emission of gasses, which is listed as one of the factors contributing to global warming, though India's per capita emissions remain lower than in advanced economies (World Bank, 2021).

The Indian perspective towards carbon emissions and renewable energy is coloured by its development goals and how the country seeks to balance economic development with environmental conservation. Under the framework of the Paris Agreement, India has submitted its intended nationally determined contributions, enhancing the emission intensity of GDP by 33-35% by 2030 from emissions in 2005. It also intends that 40% of its installed power capacity will come from non-fossil fuel-based generation sources by the year 2030, which shows its concern towards global warming (IRENA, 2019).

2. Norway

Norway has a large amount of oil and gas exportation but has a comparatively small CO₂ emission rate for a developed nation. This is mainly because Norway utilizes more than 95% of renewable hydropower in electricity generation (Norwegian Ministry of Climate and Environment, 2021). Most of Norway's CO₂ emissions originate from oil and gas industries, transportation, and a few other industries. But, in terms of the total emissions in absolute terms, it accounts for less than 1% of global carbon emissions (World Bank, 2021). For the same reason, it ranks among the lowest in Europe regarding cumulative CO₂ emissions per capita due to the use of clean energy sources for power plants and also due to awareness among the citizens.

Norway takes active, globally leading climate action. Though a major fossil fuel exporter, Norway has pledged to reduce its domestic carbon emissions by at least 55% below 1990 levels by 2030, in line with Europe's climate ambitions (Norwegian Ministry of Climate and Environment, 2021). The Norwegian government has set a target to become carbon-neutral by 2050, with key focuses on curbing emissions across the oil and gas industry, using carbon capture and storage (CCS) technology, and moving to electric transportation. Norway's renewable energy policies are advanced compared with other countries. As a leader in electric vehicle (EV) adoption, Norway has implemented incentives such as tax breaks, toll exemptions, and free public parking to promote EV use. In recent years, EVs have made up over 80% of new car sales in Norway. Norway is also investing in offshore wind energy, including offshore floating wind farms, and exploring green hydrogen for shipping and heavy industry, in addition to its extensive hydropower resources.

II. LITERATURE REVIEW

Discussing the relationship between renewable energy adoption and CO₂ emissions we have referenced two papers.

Subhash Kumar and Reinhard Madlener's "CO₂ Emission Reduction Potential Assessment Using Renewable Energy in India" uses the LEAP energy model to analyze India's ability to reduce CO₂ emissions through increasing renewable energy consumption. The report recognizes India's rapid economic growth and rising energy demand, as well as its significant reliance on fossil fuels. After analyzing multiple scenarios, including business-as-usual (BAU) and accelerated renewable energy technology (ARET) scenarios, the authors conclude that greater renewable energy adoption can result in significant CO₂ emission reductions. The ARET scenario, which includes maximum renewable energy potential and energy efficiency gains, predicts a 74% reduction in CO₂ emissions by 2050, with renewables accounting for 36% of electricity output. This is in stark contrast to the BAU scenario, which predicts continued high reliance on fossil fuels and negligible reduction in CO₂ emissions. The study examines several renewable energy sources (big hydro, small hydro, wind, solar, and biomass) and emphasizes the necessity of resolving cost, technology maturity, and implementation difficulties for each. The authors believe that a deliberate transition to renewable energy, combined with energy efficiency measures, is critical for reducing India's CO₂ emissions and guaranteeing long-term energy security.

"The dynamic impact of renewable energy and institutions on economic output and CO₂ emissions across regions" by Bhattacharya, Awaworyi Churchill, and Paramati uses panel data from 85 countries (1991-2012) to analyze the impact of renewable energy and institutional quality on economic growth and CO₂ emissions. Employing system GMM, the study finds that renewable energy consumption significantly boosts economic output and reduces CO₂ emissions, although effects vary across regions. Stronger institutions positively influence economic growth and decrease emissions. Heterogeneity across countries is significant. The authors conclude that both renewable energy deployment and robust institutions are vital for sustainable economic development and that policy effectiveness hinges on institutional strength.

Examining the impact of Carbon Policy on CO₂ emissions we have taken the help of two papers.

The paper "Impacts of Carbon Pricing on Developing Economies" by Koh, Jane; Shazali Johari; Ahmad Shuib et al. investigates the implications of carbon pricing mechanisms on developing economies, including carbon taxes, emissions trading systems (ETS), and results-based carbon finance (RBCF). Recognizing the limited relevance of findings from rich countries, the study combines studies on the implications in several developing countries. The authors examine the consequences on enterprises, discovering that while initial cost increases can reduce competitiveness, revenue recycling and innovation incentives can buffer these negative effects; some data even shows that carbon pricing can improve competitiveness over time. The distributional impacts on households vary greatly; while carbon pricing can be regressive in some contexts (particularly in urban areas), it can also be progressive or proportional in others, particularly in countries with large rural populations and diverse energy consumption patterns. Macroeconomic consequences are equally complex, relying substantially on policy package design and revenue recycling strategies. Some studies reveal short-term negative GDP effects that are offset by long-term positive effects, whereas others show a more neutral total impact. The article emphasizes the significance of adjusting carbon pricing plans to unique socioeconomic circumstances, and advocates for additional methods such as environmental fiscal changes to maximize beneficial effects. The

authors conclude that a well-designed method can yield several benefits, including economic growth, job creation, increased equity, and debt reduction, all of which contribute to long-term development objectives.

"Carbon pricing and the elasticity of CO₂ emissions" by Rafaty, Dolphin, and Pretis uses a novel approach to estimate the impact of carbon pricing on CO₂ emissions across five sectors in 39 OECD countries (1990-2016). The study employs a generalized synthetic control method to estimate the average treatment effect of introducing carbon pricing, finding a 1-2.5% reduction in CO₂ emissions growth. Further analysis reveals a small, imprecisely estimated semi-elasticity of emissions with respect to carbon prices. Simulations suggest that carbon pricing alone, even globally implemented, is unlikely to achieve emission reductions consistent with the Paris Agreement without complementary policies. The authors emphasize the challenges of prior research in obtaining standardized carbon pricing data and propose a novel emissions-weighted carbon price measure to overcome these challenges.

While reviewing the impact of Renewable Energy on Economic Growth we have discussed two papers.

"The Impact of Renewable Energy Consumption to Economic Growth: A Panel Data Application" by Roula Inglesi-Lotz investigates the relationship between renewable energy consumption and economic growth using panel data from OECD countries (1990-2010). The study employs a Cobb-Douglas production function and Pedroni's panel cointegration test to determine the long-run relationship. Results show a positive and statistically significant impact of renewable energy consumption (both total consumption and share of renewables in the energy mix) on GDP and GDP per capita, supporting the idea that policies promoting renewable energies benefit both environmental and economic conditions. The findings are robust across different model specifications, confirming the long-run relationship between renewable energy and economic growth.

"The impact of renewable energy consumption to economic growth: A replication and extension of Inglesi-Lotz (2016)" by Dogan, Altinoz, Madaleno, and Taskin replicates and extends Inglesi-Lotz's (2016) study on the relationship between renewable energy consumption and economic growth in OECD countries (1990-2010). Replicating the original study using OLS with fixed effects, the authors find similar positive results. However, using panel quantile regression to address heterogeneity, they find a positive impact of renewable energy on growth only for lower quantiles, while the effect is negative for higher quantiles when renewable energy is measured as an absolute value. When measured as a share of total energy, the impact is mostly negative. These findings contrast with the original study and highlight the importance of considering the distributional effects of renewable energy on economic growth across different income levels.

While discussing the application of Kuznet Curve we have taken into account two papers.

Avik Sinha and Muhammad Shahbaz's study, "Estimation of Environmental Kuznets Curve for CO₂ Emission: Role of Renewable Energy Generation in India," looks at the Environmental Kuznets Curve (EKC) hypothesis for CO₂ emissions in India from 1971 to 2015. To overcome methodological constraints found in earlier EKC investigations, the authors use an Autoregressive Distributed Lag (ARDL) model with several structural breaks and unit root tests. Their data support an inverted U-shaped EKC, which shows that CO₂ emissions initially rise with economic expansion but then fall after a particular income threshold (about USD 2937.77).

Renewable energy generation has a large negative impact on CO₂ emissions, with long-term elasticity outweighing short-run elasticity. Trade volume has a negative association with emissions, indicating a positive technological transfer effect. While overall factor productivity has little influence, energy consumption has a favorable impact, with long-run elasticity exceeding short-run elasticity, exposing inefficiencies in India's energy system. The analysis concludes with policy suggestions that emphasize a phased transition to renewable

energy sources, including discriminatory subsidies, in order to balance economic growth and environmental sustainability. However, the EKC's turnaround point is outside of the sample period, indicating that more research is needed to adequately capture the long-term effects.

"A Kuznets Curve in Environmental Efficiency: An Application on OECD Countries" by Zaim and Taskin investigates the Environmental Kuznets Curve (EKC) hypothesis using a nonparametric approach to calculate an environmental efficiency index for OECD countries (1980-1990). The authors develop an index measuring the opportunity cost of transforming production processes to reduce undesirable outputs. Their analysis reveals an inverted U-shaped relationship between environmental efficiency and per capita GDP, supporting the EKC hypothesis. Further analysis explores factors influencing environmental efficiency, including GDP per capita, population density, R&D expenditure, and the share of manufacturing in GDP. The study finds evidence suggesting that environmental efficiency improves at lower income levels, deteriorates at intermediate levels, and then improves again at higher income levels.

III. Key Policies Implemented

Key renewable energy policies implemented in Norway after the 1990s. (Dwija)

1990s: Norway's Climate Policy and International Commitments

- **UNFCCC and Kyoto Protocol:** Norway became a signatory to the United Nations Framework Convention on Climate Change (UNFCCC) and later the Kyoto Protocol, committing to reducing its greenhouse gas emissions.
- **Green Taxes:** Norway introduced green taxes, including carbon taxes on fossil fuels, to discourage high emissions and promote cleaner energy alternatives. (EDF & IETA, 2013) (IEA, Norway, 2022)

2000s: Renewable Energy Promotion Act (2004)

- **Policy:** This act was designed to promote renewable energy generation, especially focusing on wind energy and small-scale hydropower projects.
- **Impact:** It introduced financial incentives and support mechanisms for companies and individuals to invest in renewable energy projects, helping to diversify the energy mix. This policy laid the groundwork for increased renewable energy capacity in Norway, particularly in the form of onshore wind power, by providing grants, subsidies, and regulatory support (UNFCCC, Norway's long-term low-emission strategy for 2050)

Enova SF (2001, Expanded in 2007)

- Enova SF is a state-owned entity aimed at promoting energy efficiency and renewable energy.
- **Policy:** Enova provides financial support for renewable energy projects, energy efficiency initiatives, and clean technology advancements. It supports projects across hydropower, wind, hydrogen, and electric mobility.

The Electricity Certificate Market (2012)

- Norway and Sweden jointly launched this market-based support system to increase renewable energy production.
- **Policy:** Renewable energy producers receive certificates for each megawatt-hour (MWh) of electricity they generate, which they can sell, promoting investment in renewable sources like wind and small hydropower projects.
- **Impact:** This market mechanism boosted investments in renewable energy production across Norway. (EDF & IETA, 2013)

Energy Policy White Paper (2016)

- This was a strategic document released by the Norwegian government outlining the future direction of Norway's energy sector up to 2030, focusing on maintaining Norway's leadership in renewable energy, increasing offshore wind, and ensuring a balanced energy transition.
- **Impact:** The paper reaffirmed the country's commitment to renewable energy and outlined strategies for reducing emissions while maintaining Norway's role as an energy exporter. (IEA, Norway, 2022)

Norwegian Climate Act (2017)

- **Policy:** The Climate Act legally binds Norway to reduce its greenhouse gas emissions by 40% from 1990 levels by 2030, and by 90-95% by 2050.
- **Impact:** This act reinforced Norway's commitment to cutting emissions and developing renewable energy infrastructure, including hydropower, wind, and offshore wind projects. (EDF & IETA, 2013) (IEA, Norway, 2022).

Offshore Wind Power Development (2019)

- **Context:** Norway started developing offshore wind power, with a particular focus on floating wind farms in the North Sea.
- **Policy:** The government opened two areas, Utsira Nord and Sørliche Nordsjø II, for offshore wind projects.
- **Impact:** Offshore wind is expected to become a significant part of Norway's renewable energy strategy, complementing its dominant hydropower capacity (IEA, Norway, 2022).

Norwegian Green Tax Shift (2020 onwards)

- **Policy:** Norway introduced a green tax shift to promote the transition to renewable energy. This included higher carbon taxes and continued incentives for electric vehicles (EVs).
- **Impact:** Norway is a global leader in EV adoption, with over 50% of new car sales being electric. This shift is part of a broader strategy to reduce reliance on fossil fuels and cut emissions (EDF & IETA, 2013).

Long-Term Low-Emission Strategy (2020)

- **Policy:** Norway submitted its long-term strategy to the UNFCCC, targeting a reduction in greenhouse gas emissions by 90-95% by 2050.
- **Impact:** The strategy highlights the importance of renewable energy, electrification of various sectors, carbon capture and storage (CCS), and expanding offshore wind to meet emission targets (IEA, Norway, 2022).

2021 Offshore Wind Power Auction

- **Policy:** Norway launched its first auction for offshore wind projects, focusing on floating wind farms.
- **Impact:** The auction represents a key step in Norway's transition towards renewable energy, aiming to bolster its offshore wind capacity (EDF & IETA, 2013) (IEA, Norway, 2022).

Key renewable energy policies implemented in India

The Indian government has implemented a wide range of policies to promote renewable energy development. These policies have evolved over time, reflecting changes in technology, costs, and policy priorities.

National Action Plan on Climate Change (NAPCC), 2008:

This overarching plan established eight national missions, including the National Solar Mission, National Mission for Enhanced Energy Efficiency, and the National Mission on Sustainable Habitat. These missions provide the framework for many specific renewable energy policies.

National Solar Mission (NSM), 2010:

A flagship program aimed at significantly expanding solar power capacity. It includes various initiatives like:

- Feed-in tariffs (FITs) and Viability Gap Funding (VGF): Financial incentives to attract investment in solar projects.
- Rooftop Solar Program: Encouraging the installation of solar panels on residential and commercial buildings.
- Solar Parks: Development of large-scale solar parks to facilitate grid connectivity and reduce project development costs.
- Ultra Mega Solar Power Projects (UMSPPs): Promoting the development of extremely large solar power projects (typically exceeding 500 MW).

National Mission for Enhanced Energy Efficiency (NMEEE), 2009:

Focuses on improving energy efficiency across various sectors, indirectly contributing to lower energy consumption and reduced reliance on fossil fuels. It includes schemes like:

- Perform, Achieve and Trade (PAT) scheme: A market-based mechanism where designated large energy consumers must meet energy saving targets or trade their energy savings credits.
- Energy Efficiency Financing Platform: Facilitating access to financing for energy efficiency projects.
- National Mission on Sustainable Habitat (NMSH), 2009: Aims to make urban areas more sustainable, including by promoting renewable energy sources for buildings.
- National Wind-Solar Hybrid Policy, 2019: Aims to promote the integration of wind and solar power generation at the same location to enhance grid reliability and energy security.
- Renewable Purchase Obligations (RPOs): Electricity distribution companies are mandated to source a certain percentage of their electricity from renewable energy sources. This is a crucial policy tool that creates demand for renewable energy. The RPOs have been progressively increased over time.

IV. OBJECTIVES

1. Impact on Renewable Energy Productions on CO2 Emissions
2. Impact on Carbon Pricing Policy on CO2 Emissions
3. Impact of Renewable Energy on Economic growth
4. Application of Environmental Kuznets Curves

V. METHODOLOGY

1. Impact of Renewable Energy Production on CO2 Emissions

In the following analysis, we will assess whether an increase in renewable energy production results in a reduction of CO2 emissions. By analyzing the relation between renewable energy output and emissions data, we aim to evaluate the effectiveness of renewable energy policies in curbing carbon emissions.

The variables under study are as follows -

1. CO2 emissions per capita (Global Carbon Budget, 2023)
2. Renewable Energy Output as a measure of total energy output (Global Carbon Budget, 2023).

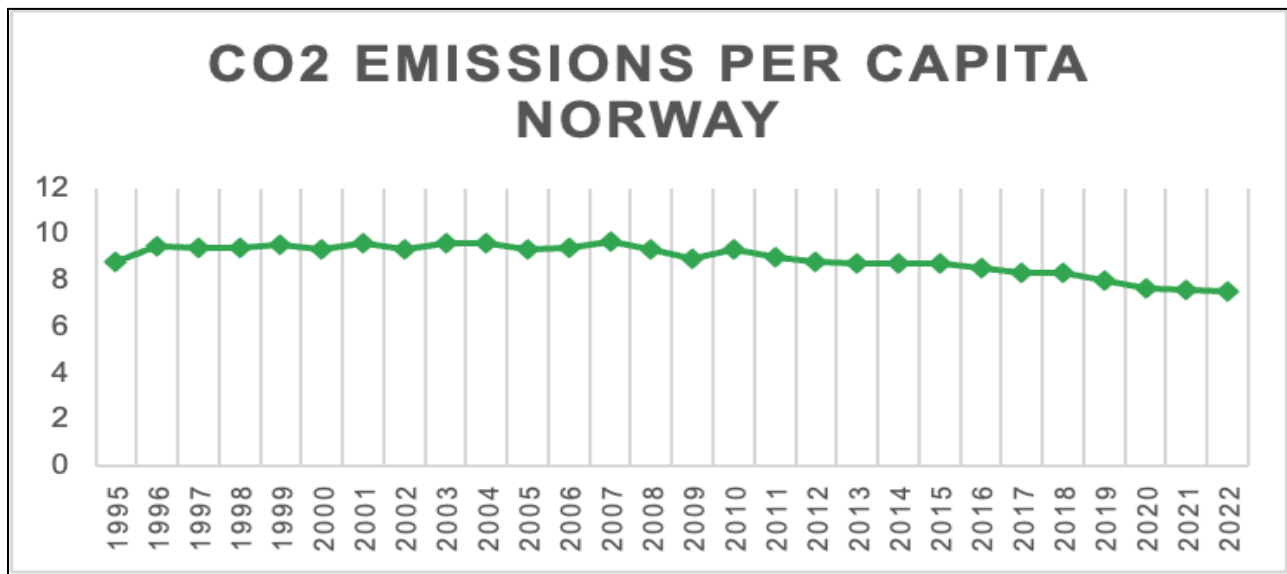
The regressions equation to test the relationship between Renewable Energy and CO2 emissions

$$CO2_{ci} = B_0 + B_1 (RE\ Share)_{ci} + u_{ci}$$

Norway

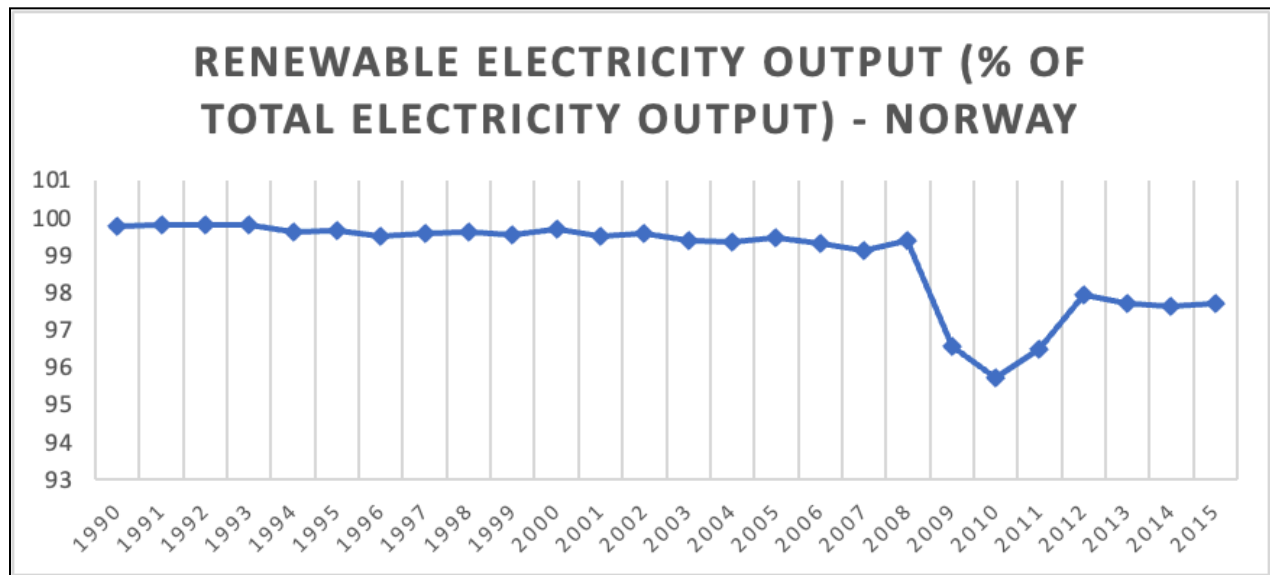
<i>Statistic</i>	<i>NOR_CO2</i>	<i>NOR_RE</i>
Mean	8.95	98.90
Standard Error	0.12	0.23
Median	9.18	99.49
Standard Deviation	0.65	1.19
Sample Variance	0.43	1.42
Kurtosis	-0.11	1.07
Skewness	-0.93	-1.49

1. CO2 emissions per capita - 1995 - 2022 (Global Carbon Budget, 2023)



- **Trend Analysis:** The graph shows Norway's CO2 emissions per capita from around 1995 to 2022. The emissions hover between 9 and 11 tons per capita, with a slight downward trend after 2016. The peak was just under 12 tons in earlier years, gradually falling to approximately 9.5 tons by 2022.
- **Inference:** Norway has maintained relatively high but stable CO2 emissions per capita, though the slight decline in recent years may be attributed to climate policies aimed at reducing carbon emissions. These could include stricter emissions standards, increased taxation on high-emission activities, and investment in cleaner technologies.
- **Descriptive Statistics**
 - Mean (8.95): On average, Norway's CO2 emissions per capita are approximately 8.95 tons over the period measured. This suggests a relatively high but stable level of emissions for an advanced economy like Norway.
 - Standard Error (0.12): The small standard error indicates the average CO2 emissions estimate is reliable, with low variability around the mean.
 - Median (9.18): The median CO2 emissions per capita are 9.18 tons, which is slightly higher than the mean, suggesting that the distribution of emissions data is skewed slightly to the left (a small proportion of lower values).
 - Standard Deviation (0.65): Emissions vary by approximately 0.65 tons from the mean on average, indicating moderate consistency in emissions over time.
 - Sample Variance (0.43): Variance of 0.43 indicates the overall spread or dispersion of emissions per capita values around the mean is relatively low.
 - Kurtosis (-0.11): A kurtosis of -0.11 suggests that the data distribution is slightly platykurtic (flatter than a normal distribution), meaning there are fewer extreme deviations.
 - Skewness (-0.93): The skewness value of -0.93 indicates the data distribution is negatively skewed. This means there is a tail on the left side of the distribution, with more instances of lower CO2 emission values than higher.

2. Renewable Energy Policy Output as a measure of total energy output (Global Carbon Budget, 2023).



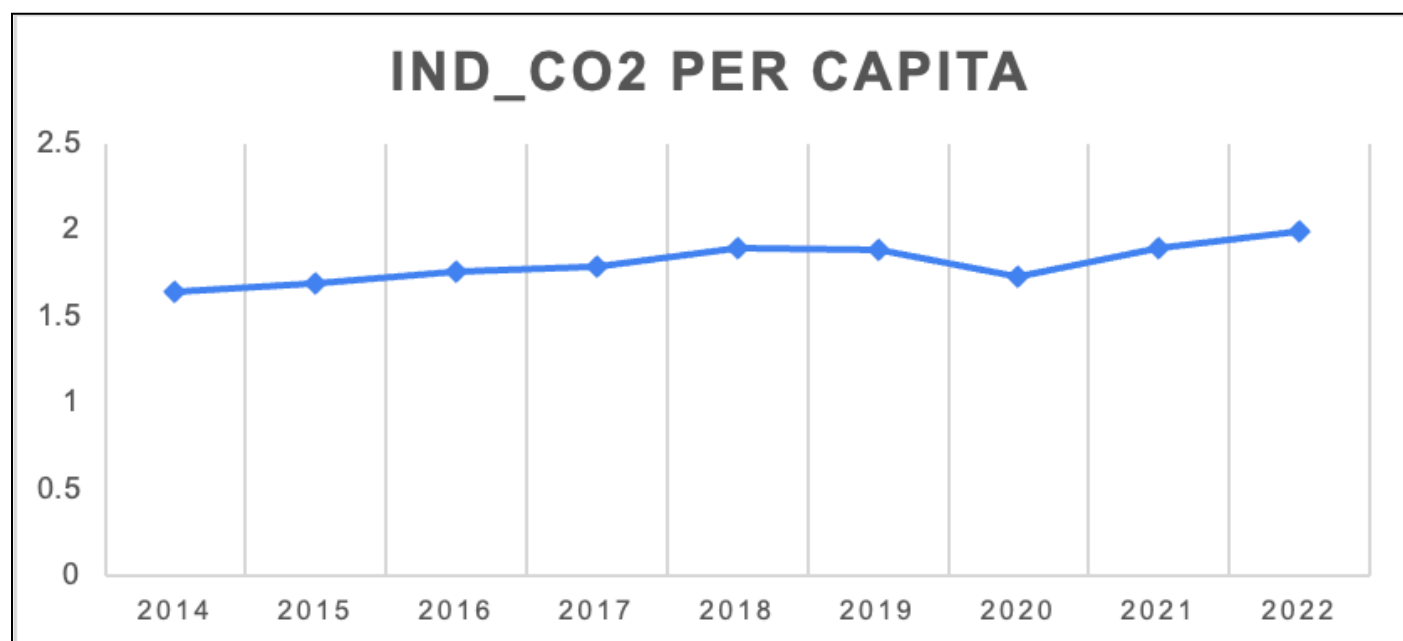
- **Trend:** Norway's renewable electricity output remains remarkably consistent and high, staying close to or above 95% for most of the observed years. The trend shows a minor dip in the middle of the timeframe, where the percentage temporarily drops to around 94%, but quickly recovers back to the upper range. This small fluctuation is likely due to short-term issues in energy production or temporary increases in non-renewable electricity usage. Overall, the graph demonstrates Norway's strong commitment to maintaining a high level of renewable energy in its electricity mix.
- **Inference:** The near-constant high percentage of renewable electricity output in Norway highlights the country's advanced stage in its transition to renewable energy. The slight dip observed does not significantly affect the overall trend, reinforcing Norway's position as a leader in renewable energy adoption. This consistency is likely due to a combination of factors such as a favorable geographical landscape for hydropower, well-developed infrastructure, and strong policy support for renewable energy. Norway's approach serves as a model for other countries aiming to achieve similar levels of renewable energy integration.
- **Descriptive statistics**
 - Mean (98.90%): On average, nearly 99% of Norway's total energy comes from renewable sources, indicating an extremely high penetration of renewable energy in the national energy mix.
 - Standard Error (0.23): A low standard error suggests a high degree of confidence that the sample mean accurately reflects the true population mean for renewable energy output.
 - Median (99.49%): Half the data is above this value, which is slightly higher than the mean. This indicates that most of Norway's energy output is even closer to 100% renewable.
 - Standard Deviation (1.19): There's a slight variation in renewable energy output, though it's still a small range considering the scale, with most values close to 99%.
 - Sample Variance (1.42): Like the standard deviation, this reflects a small spread, confirming that the share of renewables is consistently very high.
 - Kurtosis (1.07): Positive kurtosis suggests the distribution is a bit more peaked than a normal distribution, meaning most data points are tightly clustered around the mean.

- Skewness (-1.49): The negative skewness indicates a left skew, meaning there are more data points slightly below the average than above. In this case, some periods or regions might report slightly lower renewable energy percentages, but these are relatively rare.

India

Statistic	IND_CO2	IND_RE
Mean	251.45	16.88
Standard Error	7.51	0.57
Median	250.98	16.38
Standard Deviation	23.75	2.91
Sample Variance	564.20	8.48
Kurtosis	-1.03	1.02
Skewness	-0.02	0.95

1. CO2 emissions per capita - 2014 - 2022 (Global Carbon Budget, 2023)

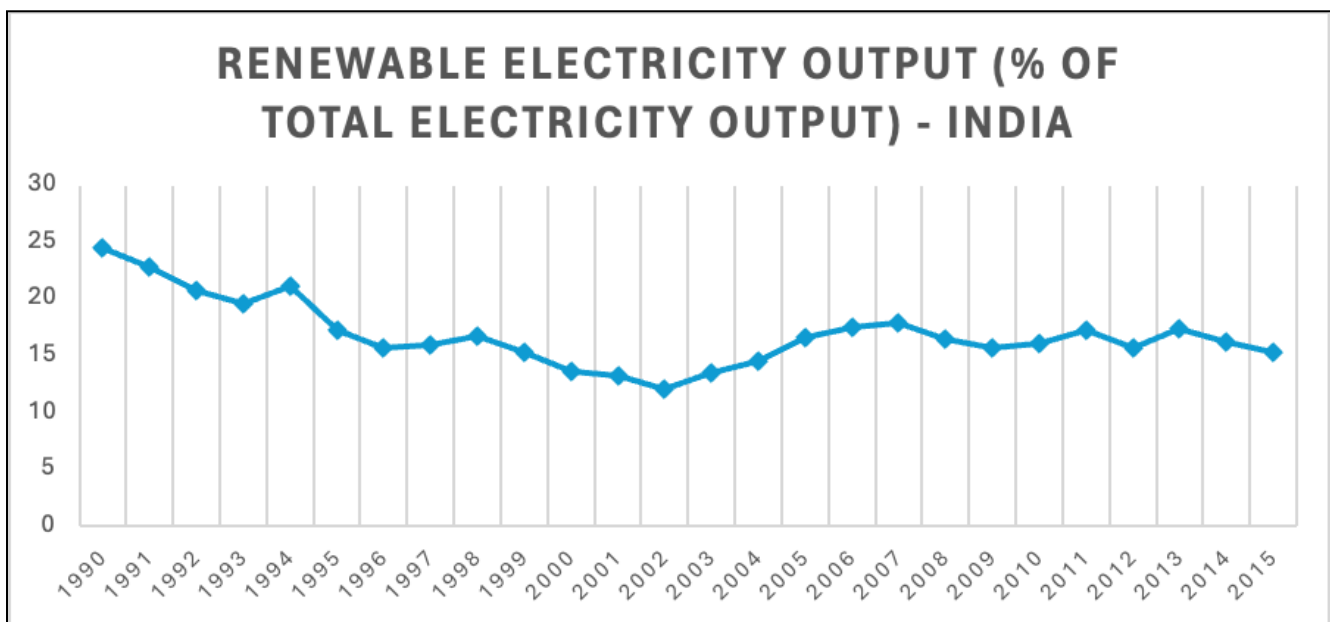


- **Trend Analysis:** - From 2014 to 2022, CO2 emissions per capita in India show a consistent upward trend. The emissions started at around 1.5 tons per capita in 2014 and gradually increased to about 2.5 tons per capita by 2022. There are no sudden spikes or dips; rather, the increase is steady across the years.
- **Inference:** - The steady increase in emissions can be attributed to industrial growth, urbanization, and increased energy consumption as India's economy expanded. The lack of sharp rises or falls indicates that India's emissions growth is a long-term trend, which could be challenging to reverse without significant interventions.

● Descriptive Statistics

- Mean (251.45): On average, India's CO2 emissions per capita, as measured in tons per capita, are 251.45 over the period measured.
- Standard Error (7.51): The small standard error indicates the CO2 emissions data is fairly consistent, with minimal fluctuation around the mean.
- Median (250.98): The median value is 250.98, very close to the mean, which suggests a symmetrical distribution of data with no significant outliers.
- Standard Deviation (23.75): Emissions deviate by 23.75 units from the mean on average, which shows moderate variability in CO2 emissions over time.
- Sample Variance (564.20): This variance indicates some dispersion in the dataset, meaning there are occasional differences between years.
- Kurtosis (-1.03): The negative kurtosis value indicates the data is platykurtic (flatter than normal), meaning there are fewer extreme values (both high and low) in the dataset.
- Skewness (-0.02): The skewness is nearly zero, implying that the distribution of CO2 emissions is symmetrical, with no significant skew toward higher or lower values.

2. Renewable Energy Policy Output as a measure of total energy output (Global Carbon Budget, 2023).



- **Trend:** The renewable electricity output in India shows a fluctuating trend over the years, generally ranging between 10% and 25%. Initially, there is a slow decline from the higher values in the early years, reaching a low point around the middle of the observed period. After that, a gradual increase is seen, indicating a renewed focus on boosting renewable energy sources towards the latter part of the timeframe. This pattern suggests a non-linear development in India's adoption of renewable energy, where progress has been inconsistent and possibly influenced by external factors like policy changes, economic conditions.
- **Inference:** The variability in India's renewable electricity output points to the country's challenges in maintaining a consistent transition to renewable energy. Factors such as infrastructure limitations, investments in fossil fuel-based power, and varying government policies might be influencing this uneven growth. While the recent upward trend is encouraging,

it indicates that India is still in the process of strengthening its renewable energy capabilities. This suggests a need for more stable policies and increased investment in green technology to sustain the growth in renewable energy output.

- **Descriptive Statistics**

- Mean (16.88%): On average, renewable energy accounts for 16.88% of India's total energy output. This suggests that, while renewable energy sources are a growing part of India's energy mix, they still represent a relatively small portion overall compared to fossil fuels.
- Standard Error (0.57): A small standard error indicates that the mean is a good representation of the true value, with little variability between samples.
- Median (16.38%): The median being slightly lower than the mean suggests a slight positive skew in the data, where some values above the average are pulling the mean up.
- Standard Deviation (2.91%): This shows a moderate variation in the share of renewable energy output, meaning the percentage of renewables can vary by about 2.91% from the mean across the dataset.
- Sample Variance (8.48): The variance confirms that the percentage of renewable energy in India's total output can fluctuate moderately.
- Kurtosis (1.02): Positive kurtosis indicates a slightly peaked distribution, meaning most data points are closer to the mean, with few extreme outliers.
- Skewness (0.95): Positive skewness suggests that the distribution of renewable energy output is skewed to the right, with a few higher-than-average values pulling the data upwards. This could indicate periods or regions where renewable energy made a significantly higher contribution than usual.

2. Impact of Carbon Pricing on CO2 Emissions

The following analysis will look at the effectiveness of carbon pricing mechanisms, such as carbon taxes and subsidies, in reducing CO2 emissions per capita. Carbon pricing aims to internalize the external costs of carbon emissions, providing financial incentives for both producers and consumers to reduce their carbon footprints. First, we study the descriptive characteristics of the data to understand their general trend, and then run regression analysis to understand the impact of the specific policy tool on CO2 emissions per capita. The double log functional form of the model is chosen to better reflect the relative change of CO2 emissions in response to policy tools -

$$\text{Log}(CO_2 \text{ Emissions}) = \log \beta_0 + \log \beta_1 (\text{Aggregate tax/subsidy collection}) + ui$$

Norway

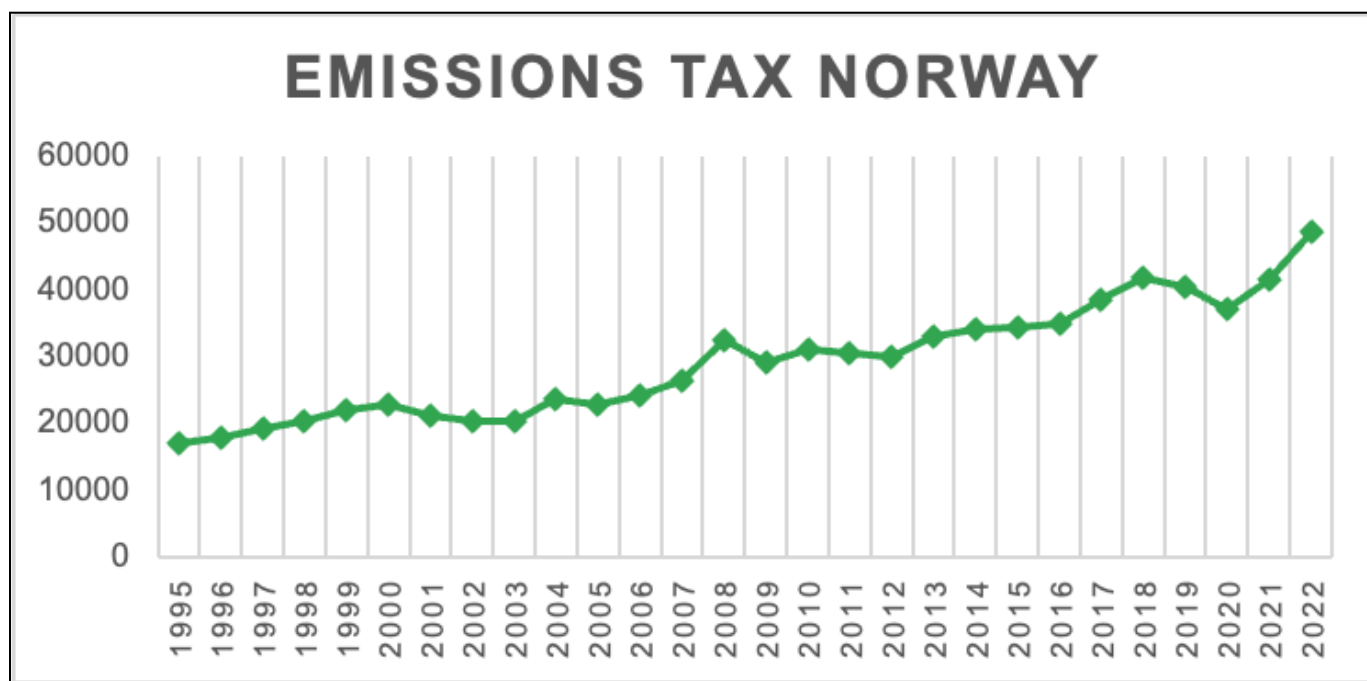
To evaluate the influence of renewable energy policies, notably carbon pricing, on CO2 emissions, we will use tax and subsidy data from Norway. Norway's comprehensive carbon pricing mechanisms, along with large subsidies to promote renewable energy, provide a perfect framework for analyzing the consequences of the policies. By studying data on carbon-related taxes and renewable energy subsidies,

we can assess how efficiently these measures cut CO2 emissions and incentivise cleaner energy generation, providing insights into the larger implications of carbon pricing for emission reductions

The variables under study are as follows - CO2 emissions per capita, Total Taxes collected and Total Subsidies disbursed.

<i>Statistic</i>	<i>NOR_CO2</i>	<i>NOR_TAX</i>	<i>NOR_CO2_SUBS</i>
Mean	8.95	29139.61	8.23047122
Standard Error	0.12	1607.51	0.15860466
Median	9.18	29676.00	8.36379025
Standard Deviation	0.65	8506.12	0.50155197
Sample Variance	0.43	#####	0.25155438
Kurtosis	-0.11	-0.68	-1.6348917
Skewness	-0.93	0.44	-0.3873816

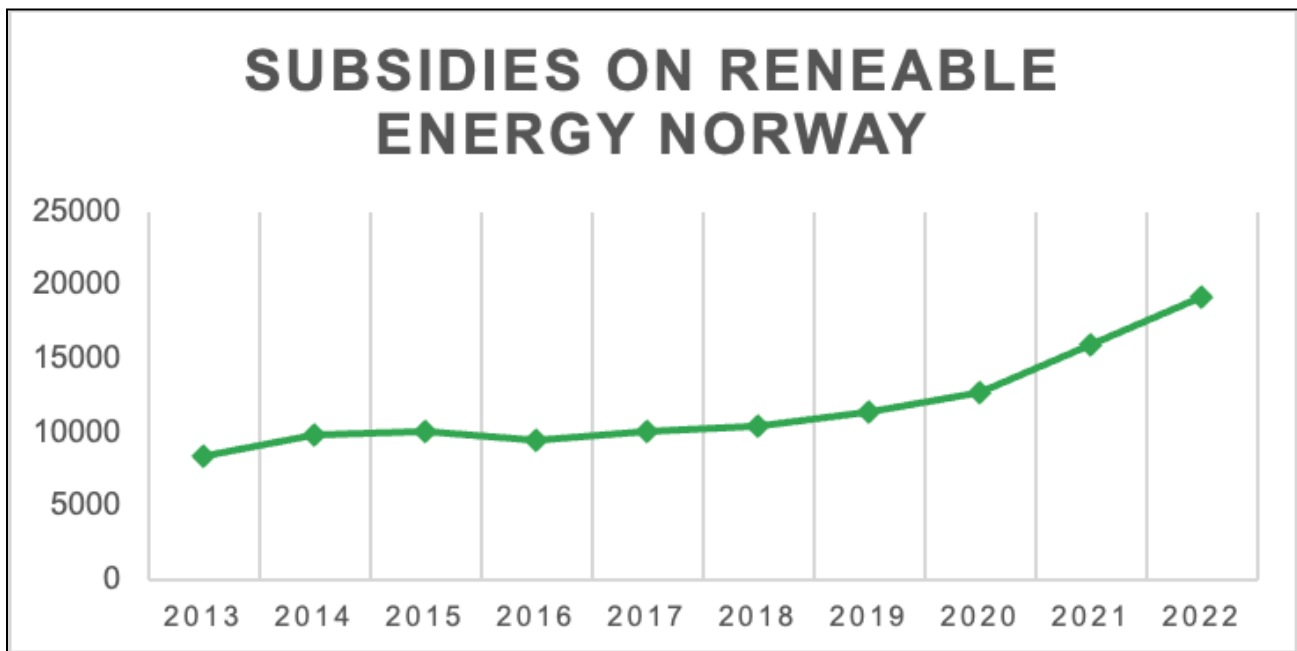
1. Total Environmental Tax collection - 1995 - 2022 (Statistics Norway, 2024)



- **Trend Analysis:** The environmental tax revenue shows a steady increase from 2013, starting at around 30,000 million NOK, and rises to almost 60,000 million NOK by 2022. The graph indicates that the tax collection has almost doubled over the period.
- **Inference:** The upward trend reflects Norway's aggressive use of environmental taxation as a policy tool. This could include taxes on carbon emissions, pollution, fossil fuel use, and other environmentally harmful activities. The increased taxation suggests the government is attempting to make environmentally unfriendly behaviors financially unattractive.
- **Descriptive Statistics**

- Mean (29,139.61 million NOK): Norway collected an average of 29,139.61 million NOK in environmental taxes annually.
- Standard Error (1,607.51 million NOK): The standard error is relatively high, suggesting some variability or uncertainty in the yearly tax data.
- Median (29,676 million NOK): The median environmental tax collection is 29,676 million NOK, very close to the mean, indicating a fairly symmetrical distribution of the data.
- Standard Deviation (8,506.12 million NOK): The high standard deviation of 8,506.12 million NOK shows significant variation in the amount of tax collected year to year.
- Sample Variance (Not shown): The exact value is missing due to an error, but it would reflect the large spread in the tax data.
- Kurtosis (-0.68): The negative kurtosis indicates the distribution of tax collection data is slightly flatter than normal (platykurtic).
- Skewness (0.44): The positive skewness of 0.44 suggests a slight rightward tail in the tax collection data, indicating that higher-than-average values occur more frequently than lower ones.

2. Total Environmental Subsidies disbursed - 2013 - 2022 (Statistics Norway , 2024)



- **Trend Analysis:** The graph shows a substantial and consistent rise in environmental subsidies starting around 2013, from under 10,000 million NOK to approximately 25,000 million NOK by 2022. The increase is particularly sharp from 2020 onwards.
- **Inference:** Norway has significantly ramped up financial support for environmental initiatives, especially renewable energy. This likely involves subsidies for solar, wind, and other renewable energy projects, alongside investments in electric vehicle infrastructure and green technology innovation.
- **Descriptive Statistics**

- Mean (8.23 million NOK): On average, Norway disbursed 8.23 million NOK in subsidies for environmental initiatives annually.
- Standard Error (0.16 million NOK): The small standard error indicates the subsidy data is fairly reliable and consistent.
- Median (8.36 million NOK): The median value of subsidies is 8.36 million NOK, very close to the mean, again suggesting a fairly symmetrical distribution.
- Standard Deviation (0.50 million NOK): The standard deviation of 0.50 million NOK indicates low variability, meaning that subsidy disbursements were quite consistent.
- Sample Variance (0.25): The variance of 0.25 reflects the low spread in the subsidy data around the mean.
- Kurtosis (-1.63): The highly negative kurtosis indicates the distribution of subsidies is very platykurtic (flatter than normal), suggesting that extreme values (either very low or very high subsidies) are rare.
- Skewness (-0.39): The slight negative skewness indicates the data is slightly left-skewed, meaning that lower subsidy values occur more frequently

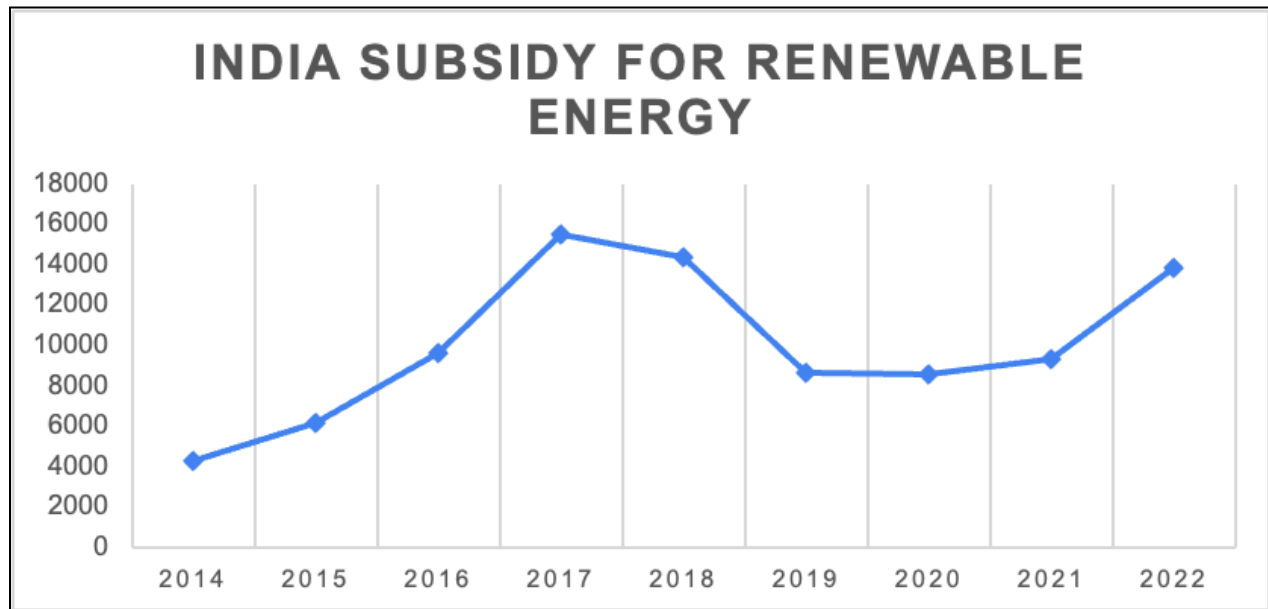
India

To estimate the influence of renewable energy regulations on CO2 emissions in India, we will examine available subsidy data from the last nine years. However, the scope of this research is limited because India does not have an explicit carbon pricing mechanism, such as carbon taxes or tradable permits. Despite this constraint, the subsidy data can still give useful information about the effectiveness of financial incentives in encouraging renewable energy and lowering emissions. The lack of direct carbon pricing makes it difficult to completely evaluate the regulatory landscape, but subsidy trends offer an essential viewpoint on India's approach to mitigating CO2 emissions.

The variables under study are as follows - CO2 emissions per capita, Total Taxes collected and Total Subsidies disbursed.

Statistic	<i>IND_CO2</i>	<i>IND_SUBS</i>
Mean	251.45	10519.65
Standard Error	7.51	1229.47
Median	250.98	9485.24
Standard Deviation	23.75	3887.93
Sample Variance	564.20	15115963.37
Kurtosis	-1.03	-1.30
Skewness	-0.02	-0.10
Count	10.00	10.00

Renewable Energy Subsidies dispersed by the center 2014 - 2022 (IISD -2022)



- **Trend:** - From 2014 to 2017, renewable energy subsidies in India show a rapid increase, peaking at around 16,000 million INR in 2017. However, after this peak, subsidies sharply declined between 2017 and 2019, falling to around 7,000 million INR. From 2020 onwards, subsidies recover, with a noticeable uptick by 2022, rising to around 12,000 million INR.
- **Inference:** The sharp increase in subsidies until 2017 reflects a period of strong government focus on promoting renewable energy. The decline in subsidies after 2017 could suggest a shift in government priorities, budget constraints, or possibly reduced emphasis on renewable energy during this period. The recovery in subsidies after 2020 indicates renewed government interest in boosting renewable energy, which is crucial given the rise in CO2 emissions.
- **Descriptive Statistics**
 - Mean (10,519.65 million INR): On average, India has disbursed 10,519.65 million INR in renewable energy subsidies per year during the period analyzed.
 - Standard Error (1,229.47 million INR): A relatively large standard error suggests there is significant variability in subsidies disbursed, meaning the subsidy values have fluctuated from year to year.
 - Median (9,485.24 million INR): The median subsidy value is 9,485.24 million INR, which is lower than the mean, indicating that there are some higher-than-average subsidy values (which increase the mean).
 - Standard Deviation (3,887.93 million INR): This large standard deviation reflects high variability in the amount of subsidies disbursed, suggesting the government's support for renewable energy fluctuates significantly from year to year.
 - Sample Variance (15,115,963.37 million INR): This very high variance underscores the large spread in subsidy data, which is consistent with the high standard deviation.
 - Kurtosis (-1.30): The kurtosis value indicates the distribution of subsidies is platykurtic, meaning there are fewer extreme subsidy values than expected in a normal distribution.
 - Skewness (-0.10): The slightly negative skewness suggests the distribution of subsidy data.

3. Impact of Renewable Energy on Economic growth

The impact of renewable energy on economic growth can be studied by analyzing the relationship between the share of renewable energy in the overall energy mix and Gross Domestic Product (GDP). By examining trends in renewable energy share alongside GDP data, we can assess whether increased investment in renewable energy correlates with economic growth. This analysis will help determine if countries that prioritize renewable energy development experience enhanced economic performance, providing insights into the potential benefits of transitioning to cleaner energy sources.

Variables taken:

1. **Renewable Electricity Output as a % of Total Electricity Output (1990 - 2015)** : This is the independent variable taken as a proxy for renewable energy, because it reflects the degree of transition to sustainable energy, which can influence economic growth through factors like energy costs, investment, and environmental impact.
2. **GDP in USD Billions (1990 - 2015)** : This is the dependent variable taken as a proxy for economic growth.

The regression that has been run tells us how an increase in renewable energy production affects the economic growth of a country.

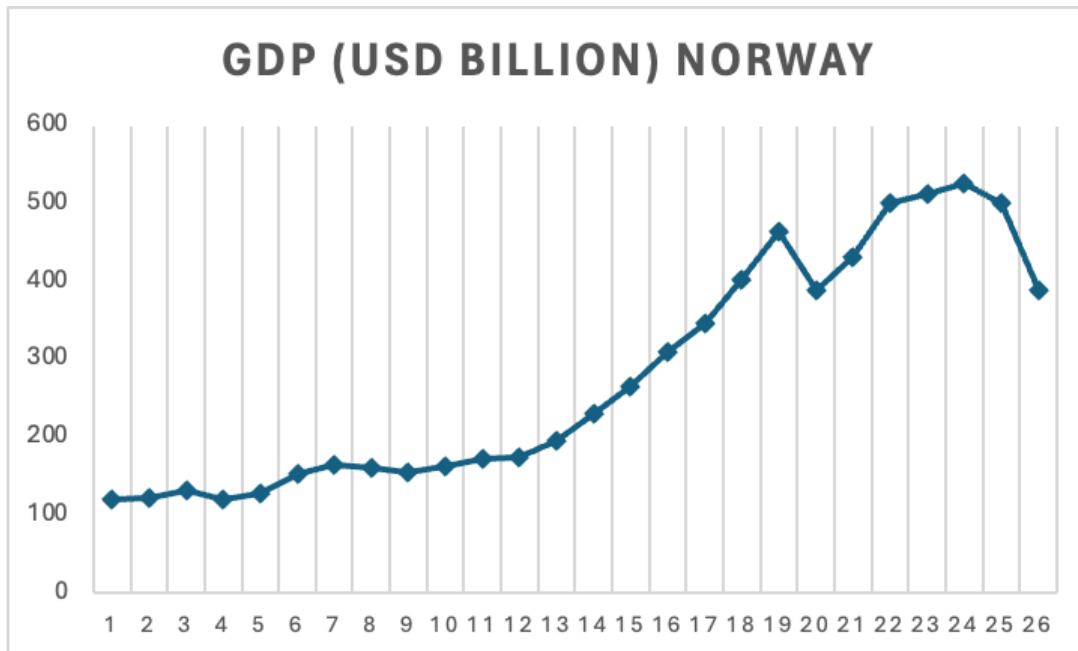
$$(GDP)_{ci} = \text{Beta}0 + \text{Beta}1 * (\text{Renewable Energy Output } \%)_{ci} + (u)_{ci}$$

Where, “c” is the country chosen and “i” is the time period of data.

Norway

<i>Statistic</i>	<i>Renewable electricity output (% of total electricity output)</i>	<i>GDP (USD Billion)</i>
Mean	98.9033	277.8281
Standard Error	0.2336	28.8692
Median	99.4856	212.6502
Standard Deviation	1.1913	147.2048
Sample Variance	1.4192	21669.2638
Kurtosis	1.0693	-1.4132
Skewness	-1.4908	0.4902

GDP in US Billions



For Norway, the graph shows a e. Around point 12 (likely early 2000s), the country's GDP rises sharply, reflecting a period of prosperity, likely tied to the booming oil industry. However, starting from point 22, the graph exhibits fluctuations and a sharp decline towards the end. This recent volatility could be attributed to factors such as a drop in oil prices, economic adjustments, or the global impact of events like the COVID-19 pandemic, which disrupted many economies. Norway's reliance on oil exports makes its GDP vulnerable to global market changes, causing the graph to show periods of growth and subsequent downturns.

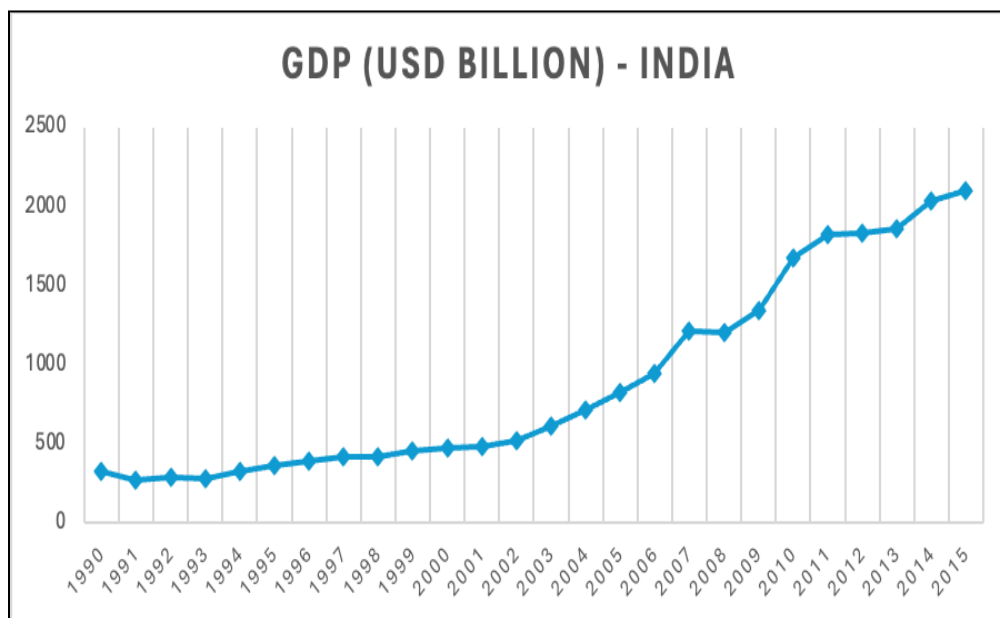
- **Descriptive Statistics**

- Mean: 277.83 billion USD - On average, Norway's GDP across the 26 years was around 277.83 billion .
- Standard Error: 28.87 billion USD - This is a larger error compared to renewable output, reflecting more variability in GDP data.
- Median: 212.65 billion USD - The median is lower than the mean, which suggests a positive skew in the data (a few larger values pulling the mean up).
- Standard Deviation: 147.20 billion USD - The high standard deviation indicates that the GDP data has a wide spread around the mean, meaning there is significant variation in Norway's GDP across the years.
- Skewness: 0.49 - A slight positive skewness indicates that the data is skewed to the right, but not strongly so.
- Kurtosis: -1.41 - A negative kurtosis (platykurtic distribution) suggests that the data has lighter tails and is more evenly spread compared to a normal distribution.

India

<i>Parameter</i>	<i>Renewable electricity output</i>	<i>GDP (USD Billion)</i>
Mean	16.88	890.9467074
Standard Error	0.57	124.1641725
Median	16.38	561.3199138
Standard Deviation	2.91	633.1155387
Sample Variance	8.48	400835.2853
Kurtosis	1.02	-0.951109327
Skewness	0.955097872	0.790232075

GDP in US Billions



India's GDP shows a consistently upward trajectory. From the early 1990s (point 1), the country's GDP increased gradually, with a significant acceleration starting around point 20 (early 2000s). This marks a period when India's economy experienced rapid expansion, driven by economic liberalization, an increase in technology and service exports, and industrial growth. Unlike Norway, India's GDP shows no significant dips throughout the timeline, indicating a steady and resilient growth pattern. The graph reflects India's status as a developing economy that has maintained momentum through structural reforms and a diversified economic base, making it less susceptible to global commodity price fluctuations than Norway.

- **Descriptive Statistics**

- **Mean (890.95 billion USD):** The average GDP over the period is 890.95 billion USD, indicating the general economic size during this time frame.

- Standard Error (124.16 billion USD): A relatively large standard error suggests there is significant fluctuation in GDP values around the mean, indicating varying growth rates over time.
- Median (561.32 billion USD): The median GDP is 561.32 billion USD, which is considerably lower than the mean, showing that higher GDP values skew the data upward.
- Standard Deviation (633.12 billion USD): This large standard deviation suggests significant variability in GDP, meaning there are notable fluctuations in the economic output over the time period analyzed.
- Sample Variance (400,835.29 billion USD): This very large variance further confirms that the GDP data is widely spread out, with some years seeing substantially different GDP figures.
- Kurtosis (-0.95): The negative kurtosis value indicates a platykurtic distribution, meaning fewer extreme values (both high and low) than in a normal distribution, with GDP being more uniformly spread out around the average.
- Skewness (0.79): A slightly positive skewness indicates that the GDP data has a moderate skew towards higher values, meaning a few years with very high GDP values likely pulled the mean up.

4. Application of Environmental Kuznets Curves

The theory of the Environmental Kuznets Curve says that as countries develop and their GDP grows, the environmental degradation will first increase and then after a turning point, it will start declining. This forms an inverted-U relationship between growth and environmental degradation. We test this theory in both countries through regression and a line graph. Variables taken are, CO2 emissions and GDP in USD Dollars, these variables have also been used before in this report.

Regression Equation:

$$(\text{Carbon Dioxide Emissions})_{ci} = \text{Beta}0 + \text{Beta}1 * (\text{GDP})_{ci} + \text{Beta}2 * (\text{GDP})_{ci}^2 + (u)_i$$

The squared GDP gives us the shape of the curve, if the coefficient of GDP square is negative, it proves the theory right.

Below are the regression results and graphs for both India and Norway.

*data taken is for 30 years, from 1990 to 2020.

Norway

```
Call:
lm(formula = COEn ~ GDPn + GDP_square)

Residuals:
    Min       1Q   Median       3Q      Max
-5342.5 -1129.6   206.2  1033.5  5565.8

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  2.706e+04  2.341e+03  11.557  3.6e-12 ***
GDPn         5.340e+01  1.805e+01   2.958  0.00623 **
GDP_square   -6.029e-02  2.915e-02  -2.068  0.04797 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2155 on 28 degrees of freedom
Multiple R-squared:  0.587,    Adjusted R-squared:  0.5575
F-statistic: 19.9 on 2 and 28 DF,  p-value: 4.195e-06
```

India

```
Call:
lm(formula = COEi ~ GDPi + GDP_square)

Residuals:
    Min       1Q   Median       3Q      Max
-148814 -31654   2781  54315 131028

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  3.996e+05  4.133e+04   9.668  2.02e-10 ***
GDPi         9.971e+02  7.859e+01  12.688  3.96e-13 ***
GDP_square   -1.002e-01  2.682e-02  -3.735  0.00085 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 80600 on 28 degrees of freedom
Multiple R-squared:  0.9846,    Adjusted R-squared:  0.9835
F-statistic: 896 on 2 and 28 DF,  p-value: < 2.2e-16
```

VI. ANALYSIS

1. Impact on Renewable Energy policy on CO2 Emissions

Norway

Parameters	Value
β_0	22.11047 (p value - 4.81e-09)
β_1	-0.19176 (P value - 3.71e-05)

$$R^2 = 0.4273 \text{ (P value - 3.706e-05)}$$

Coefficients

- **Intercept (22.11047):** The intercept represents the estimated CO2 emissions when renewable energy production is zero. It suggests that without any renewable energy production, the predicted level of CO2 emissions would be around 22.11 units. The intercept is also highly significant with a p-value of 4.81e-09.
- **NO_RE (-0.19176):** The coefficient for NO_RE is -0.19176. This indicates that for each additional unit of renewable energy production, CO2 emissions decrease by approximately 0.19176 units. The negative sign shows an inverse relationship between renewable energy production and CO2 emissions. The p-value for NO_RE is 3.71e-05, which is very small (significant at the 0.001 level). This suggests that the relationship between renewable energy production and CO2 emissions is statistically significant.

Model Fit

- **R-squared** value is 0.4273, indicating that about 42.73% of the variability in CO2 emissions can be explained by renewable energy production.
- **p-value of 3.706e-05** indicates that the model as a whole is statistically significant. This means that renewable energy production is a significant predictor of CO2 emissions in Norway.

The analysis indicates that as renewable energy production in Norway increases, CO2 emissions tend to decrease for the time period of 1990 - 2022. The relationship is statistically significant, though the model explains a moderate amount of the variation in CO2 emissions.

India

Parameters	Value
β_0	-0.31586 (p value - 0.215)
β_1	0.12160 (P value -0.215)

$R^2 = 0.1042$ (P value - 0.06685)

Coefficients

- **Intercept (0.31586):** The intercept represents the estimated CO2 emissions in India when renewable energy production is zero. It suggests that without any renewable energy production, the predicted level of CO2 emissions would be around 0.316 units. The intercept is not statistically significant with a p-value of 0.5197.
- **IND_RE (0.12160):** The coefficient for IND_RE is 0.12160. This indicates that for each additional unit of renewable energy production in India, CO2 emissions increase by approximately 0.1216 units. The positive sign suggests a direct relationship between renewable energy production and CO2 emissions in India. The p-value for IND_RE is 0.0668, which is marginally significant at the 10% level but not significant at the 5% level.

Model Fit

- **R-squared value is 0.1042**, indicating that about 10.42% of the variability in CO2 emissions in India can be explained by renewable energy production.
- **The p-value of 0.06685** indicates that the model as a whole is not statistically significant at the 5% level but is marginally significant at the 10% level. This means that renewable energy production is not a strong predictor of CO2 emissions in India.

2. Impact on Carbon pricing policy on CO2 Emissions

Norway

Efficacy of Taxes

Parameters	Value
β_0	4.31514 (p-value: 0.000000000000169)
β_1	-0.20767 (p-value: 0.000000308158557)

$$R^2 = 0.6414 \text{ (p-value: 0.0000003082)}$$

Coefficients:

- **Intercept (4.31514):** When the log of Norway's tax on emissions is 0 (meaning a baseline value for taxes), the log of per capita CO2 emissions is estimated to be 4.315. This can be thought of as a baseline level of emissions in the absence of additional tax.
- **log(NOR_TAX) coefficient (-0.20767):** This is the slope of the regression line. A 1% increase in the emissions tax is associated with a 0.2077% decrease in per capita CO2 emissions. The negative sign suggests an inverse relationship: as taxes on emissions increase, CO2 emissions decrease.
- Both the intercept and the tax coefficient are highly statistically significant (p-values < 0.001). This means there's strong evidence to reject the null hypothesis that these coefficients are zero, implying that the emissions tax has a significant effect on CO2 emissions.

Model Fit:

- **R-squared (0.6414):** About 64% of the variation in the log of per capita CO2 emissions is explained by the log of emissions taxes. This indicates a moderately strong relationship.
- **F-statistic (46.5):** This test evaluates the overall significance of the model, and the associated p-value (< 0.000001) shows the model is highly significant.

The results indicate a statistically significant and moderately strong inverse relationship between Norway's emission taxes and per capita CO2 emissions for the period 2000-2022, suggesting that increasing the tax on emissions leads to a reduction in CO2 emissions.

Efficacy Of Subsidies

Parameters	Value
β_0	4.16678 (p-value: 0.0000014)
β_1	-0.22063 (p-value:0.000238)

$$R^2 = 0.8313 \text{ (p-value: 0.0000003082)}$$

Coefficients:

- **Intercept (4.16678):** This represents the baseline level of CO2 emissions (log(NOR_CO2_SUBS)) when the log of subsidies is zero. A value of 4.166 suggests a starting point for CO2 emissions without subsidies.

- **log(NOR_SUB) coefficient (-0.22063):** This indicates that for every 1% increase in subsidies for renewable energy and environmental initiatives, there is a 0.2206% decrease in per capita CO2 emissions. The negative sign suggests that increasing these subsidies is associated with a reduction in CO2 emissions.
- Both the intercept and the subsidy coefficient are highly statistically significant (p-values < 0.001). The strong significance of the coefficients indicates that subsidies for renewable energy and environmental efforts have a meaningful and statistically significant impact on reducing CO2 emissions.

Model Fit:

- **R-squared (0.8313):** This indicates that 83.1% of the variation in the log of CO2 emissions is explained by the log of renewable energy and environmental subsidies. This is a strong model fit, implying a robust relationship between subsidies and emissions reduction.
- **F-statistic (39.43):** The high F-statistic and its very small p-value (0.000238) show that the overall model is highly significant.

The model demonstrates a strong, significant inverse relationship between subsidies for renewable energy and environmental efforts and CO2 emissions in Norway. As subsidies increase, per capita CO2 emissions decrease.

India

Efficacy of Subsidies

Parameters	Value
β_0	-0.4393 (p-value:0.2637)
β_1	0.1128 (p-value: 0.0245)

$$R^2 = 0.538 \quad (\text{p-value: } 0.0000003082)$$

Coefficients:

- **Intercept (-0.4393):** This represents the baseline level of CO2 emissions (log(IND_CO2)) when the log of subsidies (IND_SUBS) is zero. The negative value indicates a lower expected emission level without subsidies, but the intercept is not statistically significant (p-value = 0.2637), meaning it doesn't contribute much to the model's explanation of emissions.
- **log(IND_SUBS) coefficient (0.1128):** For every 1% increase in subsidies, there is a 0.1128% increase in CO2 emissions. The positive coefficient suggests that subsidies in India are associated with higher CO2 emissions, which is unexpected for environmental or renewable

energy subsidies. This coefficient is statistically significant (p-value = 0.0245), showing that the relationship is meaningful.

- The subsidy coefficient is statistically significant at the 5% level (p-value = 0.0245), while the intercept is not. The significance of the subsidy coefficient indicates a clear relationship between India's subsidies and increased CO2 emissions.

Model Fit:

- **R-squared (0.538)**: The model explains 53.8% of the variation in India's CO2 emissions. This suggests a moderate fit, with just over half of the variability in emissions being explained by subsidies.
- **F-statistic (8.152)**: The model is statistically significant overall (p-value = 0.0245), meaning that it effectively captures the relationship between India's subsidies and CO2 emissions.

The model suggests that subsidies in India (IND) are not effective in reducing CO2 emissions. In fact, the positive relationship indicates that as subsidies increase, CO2 emissions also rise. The model explains a moderate amount of the variation in emissions, and the findings highlight the inefficacy of current subsidy policies in India when it comes to curbing emissions.

3. Impact of Renewable Energy on Economic growth

Norway

Parameters	Value
β_0	9626.216346 (p value - 3.48711E-06)
β_1	-5.818095315 (P value - 5.338E-06)

$R^2 = 0.585135639$ (f stat - 33.8502331 at df = 25)

Coefficients:

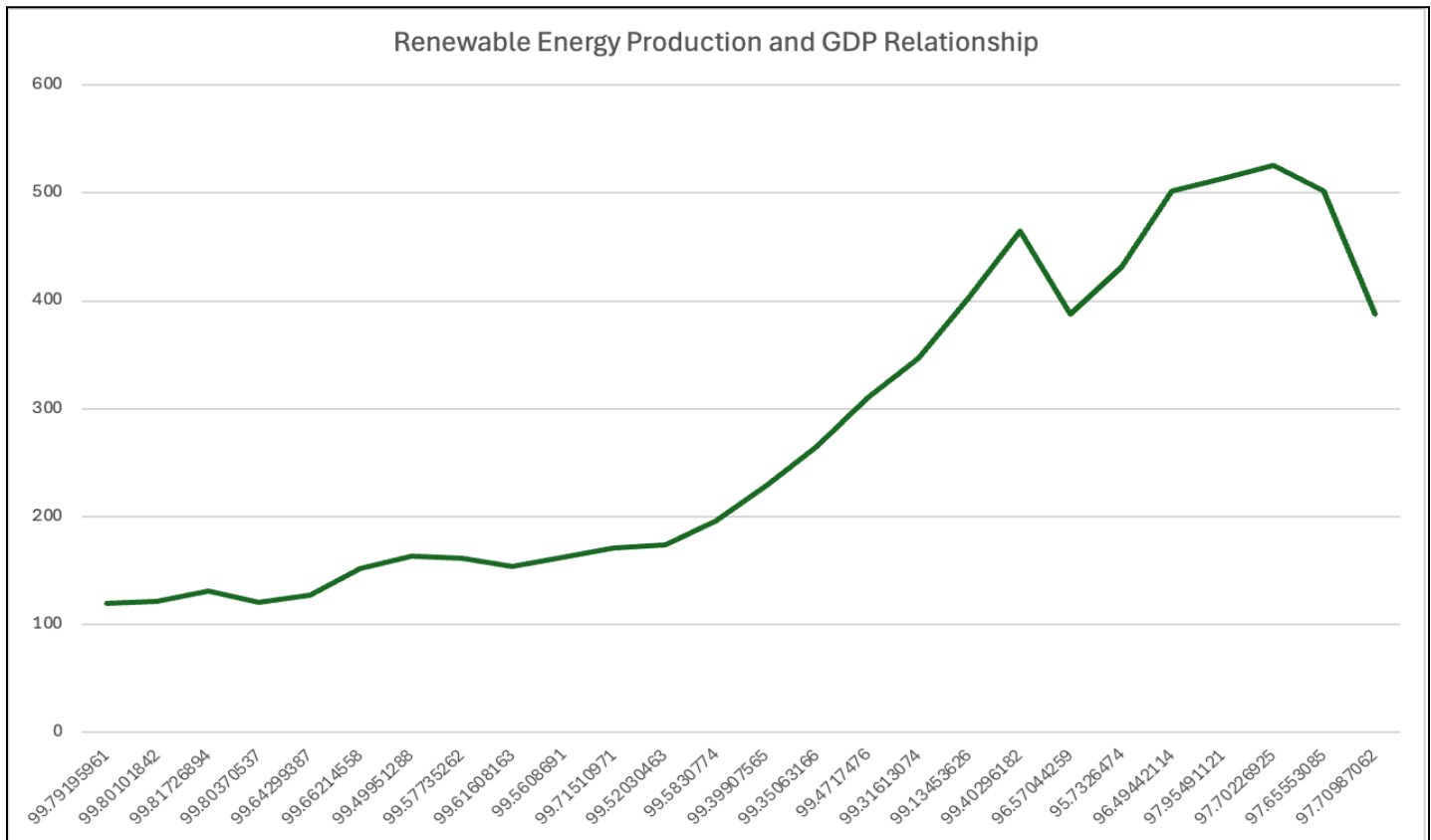
- **Intercept (9626.22)**: This represents the predicted GDP when the share of renewable electricity output is zero. The positive and significant intercept suggests that Norway has a strong baseline GDP, independent of renewable electricity. The p-value is extremely low (3.49E-06), making this a statistically significant coefficient.
- **X Variable 1 (-94.52)**: This coefficient represents the relationship between renewable electricity output and GDP. It indicates that for every 1% increase in renewable electricity output, Norway's GDP decreases by approximately 94.52 USD billion. The coefficient is statistically significant, with a p-value of 5.34E-06, meaning there is a significant inverse relationship between renewable electricity and GDP in Norway.

- **Intercept p-value (3.49E-06):** This extremely small p-value shows the intercept is highly statistically significant. **X Variable 1 p-value (5.34E-06):** The very low p-value for the renewable electricity coefficient suggests a strong, significant negative relationship between renewable electricity output and GDP.

Model Fit:

- **R-squared (0.5851):** This indicates that 58.5% of the variation in Norway's GDP is explained by the share of renewable electricity output. This is a moderate to strong fit.
- **F-statistic (33.85):** The large F-statistic and its very small p-value (5.34E-06) indicate that the overall model is highly significant.

This model demonstrates a significant inverse relationship between renewable electricity output and GDP in Norway. For every 1% increase in renewable electricity, GDP decreases by approximately 94.52 billion USD. The model explains 58.5% of the variation in Norway's GDP, which is a strong fit. However, the negative relationship suggests that Norway's GDP may be influenced by factors other than renewable energy, such as its significant oil exports, which could be affected by a transition to renewable sources..



India

Parameters	Value
β_0	1729.65088 (P value - 0.027994907)
β_1	-49.68764859 P 0.261365127

R2 = 0.052249613

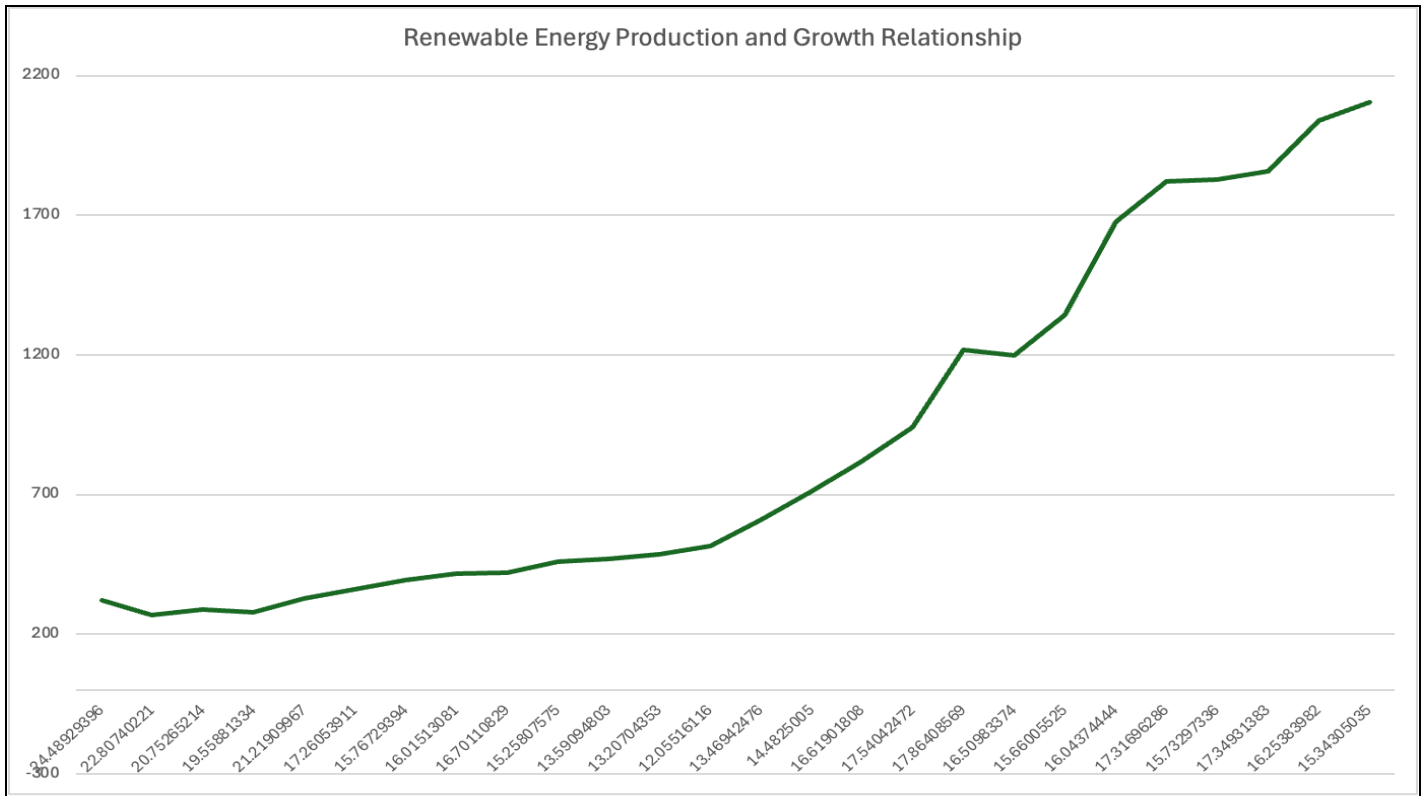
Coefficients:

- **Intercept (1729.65):** This is the predicted GDP when renewable electricity output is zero. The positive value indicates a baseline GDP level, and the intercept is statistically significant with a p-value of 0.028, meaning it contributes to the model.
- **X Variable 1 (-49.69):** This is the coefficient for renewable electricity output. It indicates that for every 1% increase in renewable electricity output, GDP decreases by approximately 49.69 USD billion. However, this coefficient is not statistically significant (p-value = 0.261), suggesting that the relationship between renewable energy output and GDP is not strong in this model.
- The **p-value for the intercept** is 0.028, which is below 0.05, indicating it is statistically significant. The **p-value for renewable electricity output** is 0.261, which is much greater than 0.05, showing that the effect of renewable electricity on GDP is not statistically significant.

Model Fit:

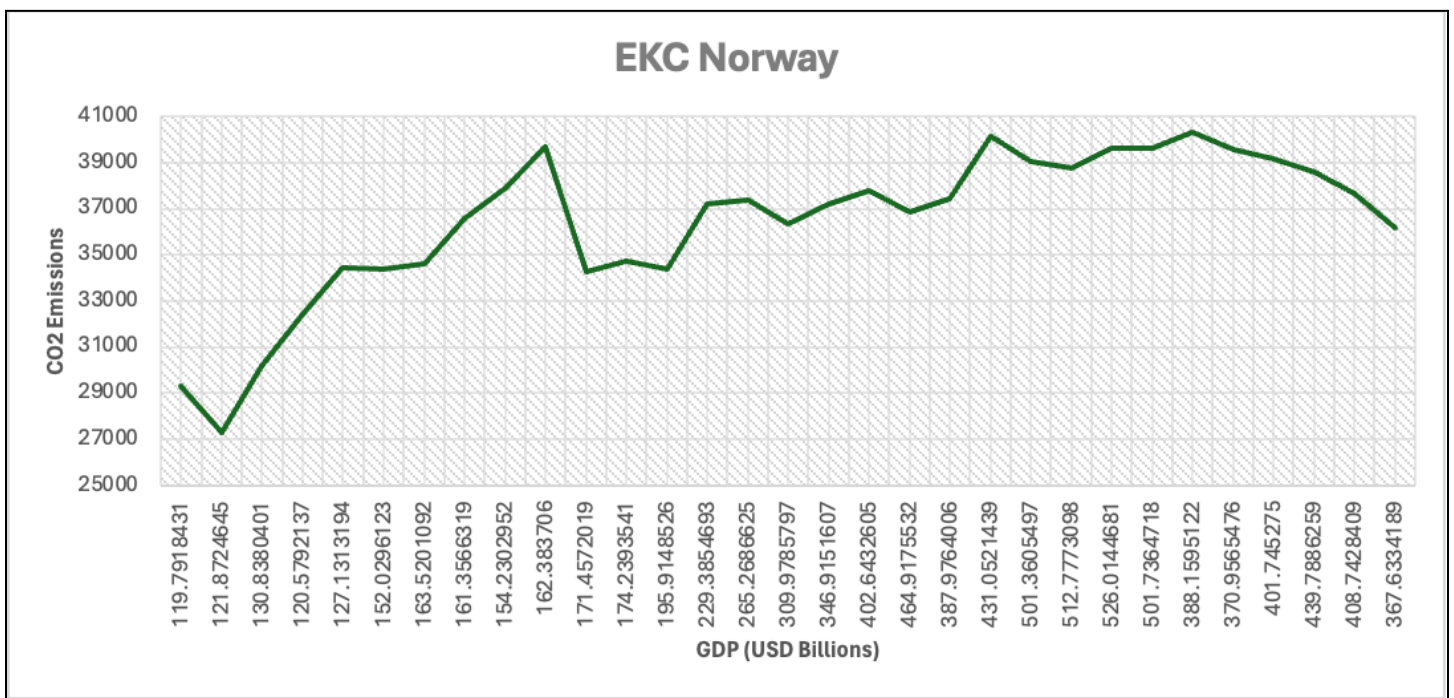
- **R-squared (0.0522):** The model explains only 5.2% of the variation in GDP, indicating a weak fit. **F-statistic (1.3231):** The F-statistic's p-value is 0.261, showing that the overall model is not statistically significant.

This model does not show a strong or statistically significant relationship between renewable electricity output and GDP in India. The low R-squared value and high p-value for renewable electricity suggest that other factors are likely more important in driving India's GDP growth than the share of renewable electricity output.



4. Application of Environmental Kuznets Curves

Norway

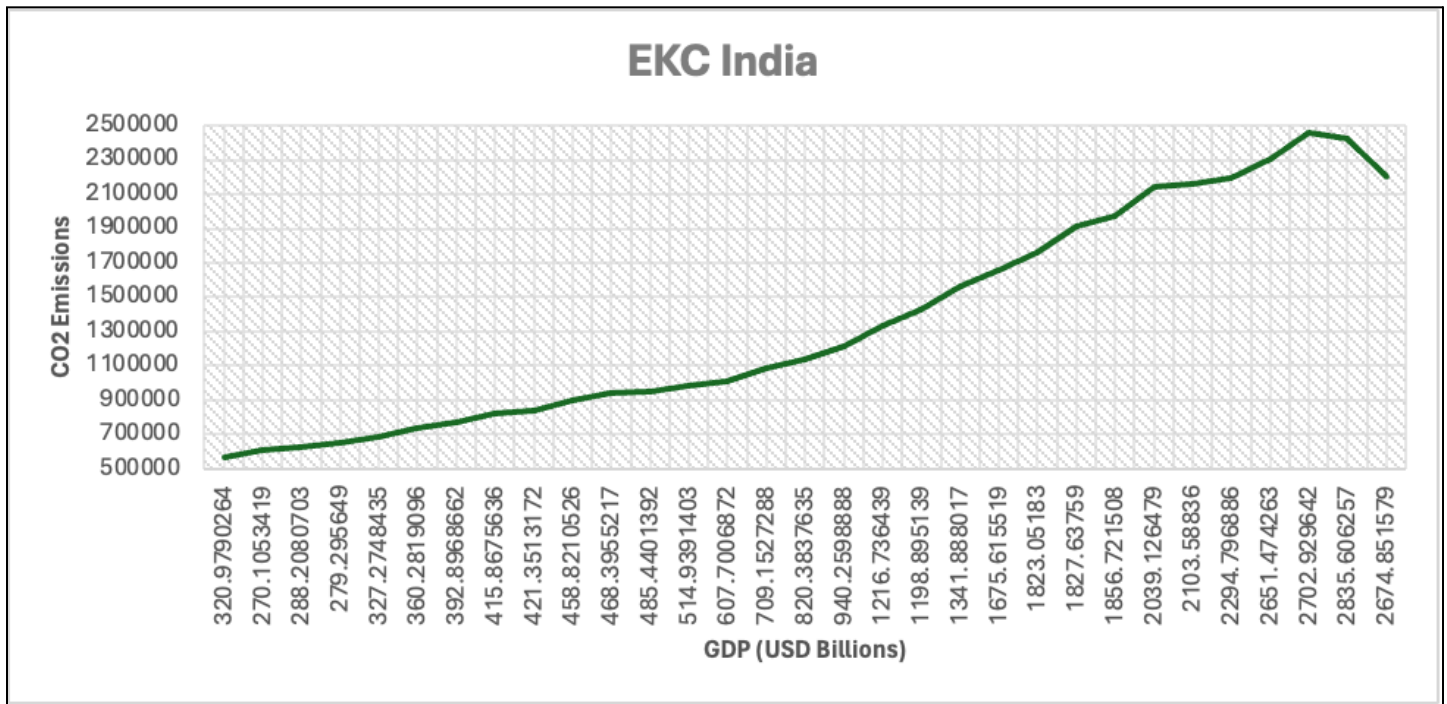


The Environmental Kuznets Curve hypothesis suggests that environmental degradation, such as CO₂ emissions, initially increases with economic growth (GDP), but after a certain level of income or GDP per capita is reached, it starts to decrease as the economy matures and invests in cleaner technologies and sustainable practices.

1. Initial Growth Stage: From the early 1990s up until the early 2000s, CO₂ emissions are rising alongside GDP. This is the typical behavior of the "growth" phase in the EKC, where economic activities and industrialization cause increased environmental degradation. As GDP grows from approximately \$120 billion to \$200 billion, CO₂ emissions increase from around 27,000 kilotons to about 38,000 kilotons.
2. Stabilization and Peak: Around 2000-2010, we notice that CO₂ emissions appear to reach a peak of approximately 40,116 kilotons, despite continuous GDP growth. This suggests the economy has reached a turning point or the "peak" of the curve, where emissions are no longer increasing significantly with GDP. In this phase, the emissions seem to fluctuate but remain within a high range, indicating that economic activities are still contributing to emissions, but efforts to stabilize or control them may have started.
3. Declining Emissions: After around 2010, CO₂ emissions began to decline even as GDP continued to rise. This decrease in emissions, coupled with continued economic growth, reflects the "declining" phase of the EKC. By 2020, emissions fall to around 36,000 kilotons, even as GDP hovers around \$360-\$440 billion USD.

This pattern aligns with Norway's strong environmental policies and transition to cleaner energy sources, such as hydropower, which contribute to the reduction in CO₂ emissions without sacrificing economic growth. The graph confirms that Norway's economy has entered the phase of sustainable growth, where further increases in GDP do not lead to corresponding increases in emissions, and the country seems to be moving towards decoupling economic growth from environmental degradation.

India



The EKC hypothesis suggests an inverted U-shaped relationship between environmental degradation (in our case CO2 emissions) and economic growth (in our case GDP). Initially, emissions rise as a country experiences economic development, but after a certain level of wealth/ income, emissions begin to decline due to factors like technological advancements, environmental regulations, and as the country shifts towards cleaner energy sources.

Based on the graph we can see that CO2 emissions in India have increased along with an increase in GDP from the time period 1990 till up to 2017-2018 approximately. After this point, we notice a slight decrease in emissions. This suggests that India might be nearing, or is in the early stages of, the turning point predicted by the EKC theory. However there are a few points to keep in mind.

- The EKC is a very simplified model. Many factors beyond GDP affect emissions such as technological innovation, energy policies and the global economic conditions. This observation of decrease in trend could be attributed to other factors.
- Since India is a developing country it faces a unique problem. India needs economic growth to improve its living standards, but this often leads to increased emissions. The Indian government has to come up with policies that balance the two.

VII. CONCLUSION

Objective	Method of study	Norway	India
Renewable Energy and Co2 emissions	Regression equation using renewable energy output and PC CO2.	As Renewable energy share goes up in total energy mix, Co2 emissions tend to fall.	Statistically Insignificant Result
Carbon pricing and Co2 emissions	Regression equation using aggregate tax/subsidy collection and PC CO2.	Taxes and both subsidies are found to reduce CO2 emissions.	Subsidies found to increase CO2 emissions. Ineffective in reducing CO2 emissions.
Renewable energy and Economic growth	Regression equation using renewable energy output and GDP.	Renewable energy output tends to reduce GDP.	Statistically Insignificant Result
Kuznets Curve	Plotting GDP and Co2 emissions.	Norway is well into its turning point - Evidence of technological effect	India at the precipice of turning point - evidence of composition effect.

The study reveals distinct differences between Norway and India in terms of renewable energy, CO₂ emissions, and economic growth, shaped by each country's unique economic and environmental context.

Norway:

1. **Renewable Energy and CO₂ Emissions:** In Norway, the increase in renewable energy's share of the energy mix shows a clear tendency to reduce CO₂ emissions, though the result is statistically insignificant. However, given Norway's high level of renewable energy, especially hydropower, the country is already in an advanced stage of its energy transition. The emissions reduction seen is likely a reflection of a mature renewable sector and well-established green policies.
2. **Carbon Pricing and CO₂ Emissions:** Carbon pricing, both through taxes and subsidies, has been somewhat effective in Norway. Taxes on carbon have played a significant role in reducing emissions, while subsidies have occasionally been found to counterintuitively increase CO₂ emissions. This indicates a need for refining how subsidies are applied to ensure they align with environmental goals. Norway's policy framework suggests that targeted taxes might be a more effective tool in emission reductions.
3. **Renewable Energy and Economic Growth:** The relationship between renewable energy output and GDP in Norway appears neutral, with renewable energy expansion not hindering growth but not substantially boosting it either. Norway's economy, being heavily resource-based, with oil

and gas playing significant roles, may explain this. Its renewable energy expansion hasn't translated into immediate GDP gains, although the long-term outlook for green technologies could bring more economic benefits.

4. **Kuznets Curve:** Norway is well past the turning point on the Kuznets Curve, experiencing a "technological effect." Advanced technologies and innovations have allowed Norway to continue economic growth while reducing CO₂ emissions. This positions Norway as a model for how developed economies can decouple economic growth from environmental degradation.

India:

1. **Renewable Energy and CO₂ Emissions:** In India, renewable energy is growing but not at a scale large enough yet to make a significant dent in CO₂ emissions. The regression results indicate a weak and statistically insignificant link between renewable energy share and emissions reduction. This may be due to the reliance on coal and other fossil fuels, which still dominate India's energy mix. India's renewable sector is growing rapidly, but its impact on emissions is not yet fully realized.
2. **Carbon Pricing and CO₂ Emissions:** Unlike Norway, India's carbon pricing policies, particularly subsidies, seem to have a counterproductive effect, sometimes leading to increased emissions. This suggests that subsidies in India might be misaligned, either supporting fossil fuel consumption indirectly or failing to incentivize clean energy investments effectively. A better design and implementation of these financial mechanisms are needed for India to reach its environmental goals.
3. **Renewable Energy and Economic Growth:** The relationship between renewable energy and GDP in India shows that renewable energy growth may hinder GDP slightly, though this result is statistically insignificant. This could indicate that while India is pushing for renewable energy, its economic structure still heavily relies on fossil fuels for industrial growth. The transition to renewable energy could initially slow growth but might bring long-term benefits as the energy market stabilizes.
4. **Kuznets Curve:** India is at the precipice of its turning point on the Kuznets Curve, where its CO₂ emissions are still rising but could soon plateau and decline as its economy transitions. Evidence of a "composition effect" is present, where economic shifts, such as the development of cleaner industries, are emerging. However, India still faces challenges due to its rapid industrialization and energy demands, requiring significant investments in green technology and policy reforms to fully reach its turning point.

Summary:

- **Norway** is in a more advanced stage of energy transition, showing evidence of a technological effect in reducing emissions while maintaining economic growth. Its policies, particularly carbon pricing, are more effective in reducing CO₂ emissions.
- **India**, on the other hand, is still grappling with the balance between economic growth and emissions reduction. Renewable energy is growing, but its effectiveness in curbing CO₂ emissions is still limited, and subsidies need to be better aligned with environmental goals. India is approaching a critical juncture in its development path where sustained investments in green energy could significantly impact both emissions and economic growth.

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